

Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained in all Python files.
2	Proper File Structure	Files and folders are logically organized	Yes	Dataset, model, utilities, and application files are organized into separate folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like <code>water_level</code> , <code>rainfall</code> , <code>flood_risk</code> , and <code>prediction_result</code> improve readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as <code>load_data()</code> , <code>preprocess_data()</code> , <code>train_model()</code> , and <code>predict_flood()</code> clearly describe their tasks.
5	Code Comments	Inline and block comments explain logic	Yes	Comments explain preprocessing, model training, and prediction workflow.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for preprocessing, training, prediction, and visualization.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common functionality is reused through functions to avoid repetition.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Handles missing datasets, invalid user inputs, and prediction errors using exception handling.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
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1	Data Preprocessing Module	Python (Pandas)	Data cleaning and feature engineering	High
2	Feature Engineering Module	Python	Data preparation before model training	High
3	Model Training Module	Python (Scikit-learn)	Training ML models	High
4	Flood Prediction Module	python	Predicting flood risk from user input	High
5	Data Visualization Module	Matplotlib	Displaying graphs and analysis	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured and organized project files.
Readability	5	Clear naming conventions and formatting
Reusability	5	Modular components reused across the project.
Documentation / Comments	4	Sufficient comments explaining major logic.
Overall Score	4.8/5	Maintainable, readable, and reusable ML projec